

SHAHZAIB SAQIB WARRAICH

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EDUCATION

University of Southern California, Viterbi School of Engineering

Los Angeles, CA

M.S., Applied Data Science

Jan 2024 - Dec 2025

EXPERIENCE

Data Scientist - Intern

Los Angeles, CA

Paramount

May 2025 - Oct 2025

- Enhanced content forecasting accuracy across multiple business KPIs by up to **30%** through the strategic integration of social media engagement signals into predictive models.
- Architected a GenAI-powered explainability framework utilizing an ensemble LLM-as-a-judge methodology with majority voting consensus to deliver granular, title-level performance insights for non-technical stakeholders, significantly improving model interpretability and business alignment.

Graduate Research Assistant

Los Angeles, CA

Data, Interpretability, Language and Learning (DILL) lab, USC-NLP Group

Jan 2024 - Dec 2025

Advisor: Swabha Swayamdipta

Mentor: Gregory Yauney

- Collaborated on projects in the intersection of LMs, Interpretability, Evaluation, Reasoning, and Alignment.
- Developed inference-time scaling frameworks to augment the factuality and reasoning ability of LMs, reducing hallucination across multiple benchmarks.
- Designed micro-benchmark meta-evaluation frameworks for reliable evaluation of micro-benchmark selection.

Research Engineer II (AI) - Full-time

Redmond, WA

Retrocausal

Dec 2022 - Dec 2023

- Designed and deployed precision assembly solutions achieving **98%** detection accuracy for Fortune 500 clients including Schneider Electric, Carrier, and Honda through advanced object detection and activity recognition algorithms (<https://retrocausal.ai/assembly-copilot/>).
- Architected an automated model training pipeline incorporating cut-paste-mix augmentation techniques to generate synthetic datasets, reducing manual data labeling requirements by **80%** while maintaining model performance standards.
- Conceptualized and led the development of a computer vision-enabled ergonomics risk assessment module for Kaizen Copilot, establishing a new product capability for workplace safety optimization (<https://retrocausal.ai/kaizen/>).

AI Engineer - Full-time

San Jose, CA

Adlytic AI

June 2021 - Dec 2022

- Directed comprehensive R&D initiatives in model optimization, implementing knowledge distillation and NVIDIA TensorRT to reduce edge inference latency by 66%, while maintaining performance within 1.5% of baseline accuracy in production environments.
- Architected and deployed a cloud-based recommender system pipeline achieving 97% accuracy, resulting in a 3x increase in product conversion rates and directly contributing to revenue growth.

INDUSTRY PROJECTS

Person Re-identification System | *PyTorch, Computer Vision, Nvidia TensorRT, FastAPI*

April 2022 - Dec 2022

- Leveraged facial recognition, super resolution, object detection, and multi-object tracking algorithms to design the model inference API. Utilized RTSP to perform object tracking on frames from live streams.
- Optimized object detection models to achieve an accuracy of up to **93%** with an IoU of nearly **0.77**.

Mobile App Recommender System | *PySpark, AWS, MongoDB, Redis, Docker*

June 2021 - Mar 2022

- Designed end-to-end system architecture on AWS (SageMaker, EMR, EC2, S3, SNS).
- Optimized matrix factorization models to generate collaborative and content based recommendations using Apache Spark.
- Stored and served recommendations using MongoDB and AWS managed Redis databases.
- Packaged and deployed application using Docker containers.
- Boosted the product's installation and client conversion rate by nearly **200%**.

RESEARCH PROJECTS AND PUBLICATIONS

- Sample, Align, Synthesize: Graph-Based Response Synthesis with ConGrs
- How Reliable is Language Model Micro-benchmarking?
- Residual balanced attention network for real-time traffic Scene Semantic Segmentation

LEADERSHIP

Viterbi Graduate Student Association (VGSA)

VP of Outreach

Aug 2024 - Present

- Representing a student body of almost 7,000 students.
- Managing internal and external partnerships.
- Planning and organizing professional career development events.

SKILLS AND INTERESTS

Skills

Languages (Python, C/C++, SQL), Deep Learning Frameworks (PyTorch, Hugging Face, vLLM), Machine Learning Frameworks (Scikit-Learn, OpenCV), Data Analysis Frameworks (Pandas, Dask, PySpark), Data Visualization (Matplotlib, Seaborn, Plotly), Cloud (AWS Sagemaker, GCP Vertex AI), Others (Docker, Kubernetes, MongoDB, Git, LaTeX)

Interests

AI Ethics and Responsibility, Hackathons, User Experience (UX) Design